Distributing Apples

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1716 | Mathematics | https://cses.fi/problemset/task/1716

There are n children and m apples that will be distributed to them. Your task is to count the number of ways this can be done.

For example, if n=3 and m=2, there are 6 ways: [0,0,2], [0,1,1], [0,2,0], [1,0,1], [1,1,0] and [2,0,0].

Input

The only input line has two integers n and m.

Output

Print the number of ways modulo 10^9+7.

Constraints

- $1 \le n,m \le 10^6$

Example

Input
3 2

Output
6